Java vs. MongoDB

Shipping Java agents to MongoDB Atlas Agent Engine

João Galego | 🧠⟩

MongoDB × Java meetup · November 2026

From atoms to agents

Education

MSc Physics



PgDip Forensics



PhD Cognitive Science (ABD)





Work

Lead ML Engineer



Solutions Architect



Head of AI



Critical software

We build systems that can't fail

🚀 Space · ✈️ Aviation · 🚆 Railway · 🏥 Medical devices · 🚗 Automotive · 🛡️ Defence · ⚡ Energy

Since 1998 · NASA · ESA · Airbus · BMW · Alstom

Want to know what I'm up to?

```
$ finger jgalego@plan.cat
Head of AI @ Critical Software.
Invited Professor @ ISEG/ULisboa.

Now:
  * bringing ML to places where "it mostly
    works" isn't enough
  * teaching (but mostly, learning)

Todo:
  - update this .plan more often
    than once a decade

Elsewhere:
  web       https://jgalego.github.io
  code      https://github.com/JGalego
  linkedin  .../in/jgalego  (fastest HELO)
```

finger me or just scan.

Once upon a time...



The kingdom

A Java agent dreams of Atlas



The curse

A dragon named `java-none`



The quest

Smuggle Java past the dragon



Happily ever after

Java answers from Atlas

Dedicated to João Mourão (MongoDB), who asked for a Disney movie

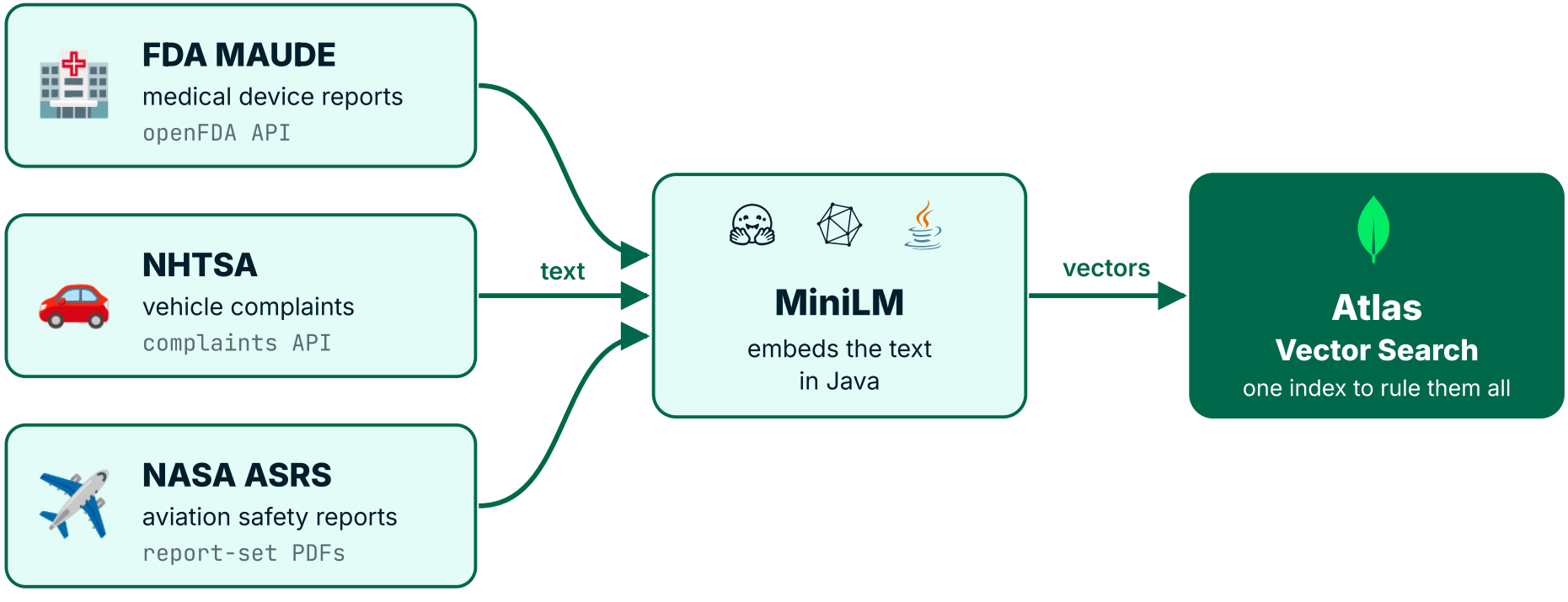
PROLOGUE

Once upon a time

near miss

noun · an incident that nearly became an accident

Where the reports come from



Look ma, no console

AtlasVectorStore.java

```
var definition = new Document("fields", List.of(
    new Document("type", "vector").append("path", "embedding")
        .append("numDimensions", dims)  // 384 MiniLM, 1024 Voyage
        .append("similarity", "cosine"),
    new Document("type", "filter").append("path", "kind"),
    new Document("type", "filter").append("path", "dataset_id")));

collection.createSearchIndexes(List.of(new SearchIndexModel(
    "near_miss", definition, SearchIndexType.vectorSearch())));
```

```
vectorSearch(fieldPath("embedding"), query, "near_miss", limit,
    approximateVectorSearchOptions(candidates)
        .filter(eq("dataset_id", "nasa-asrs")))
```

Clicks don't survive a `git clone`.

Search by meaning

🧑 "engine lost power during climb"

```
0.533  M-4, Takeoff / Launch. Pilot flying Maule M4 aircraft
       reported engine loss of power inflight…

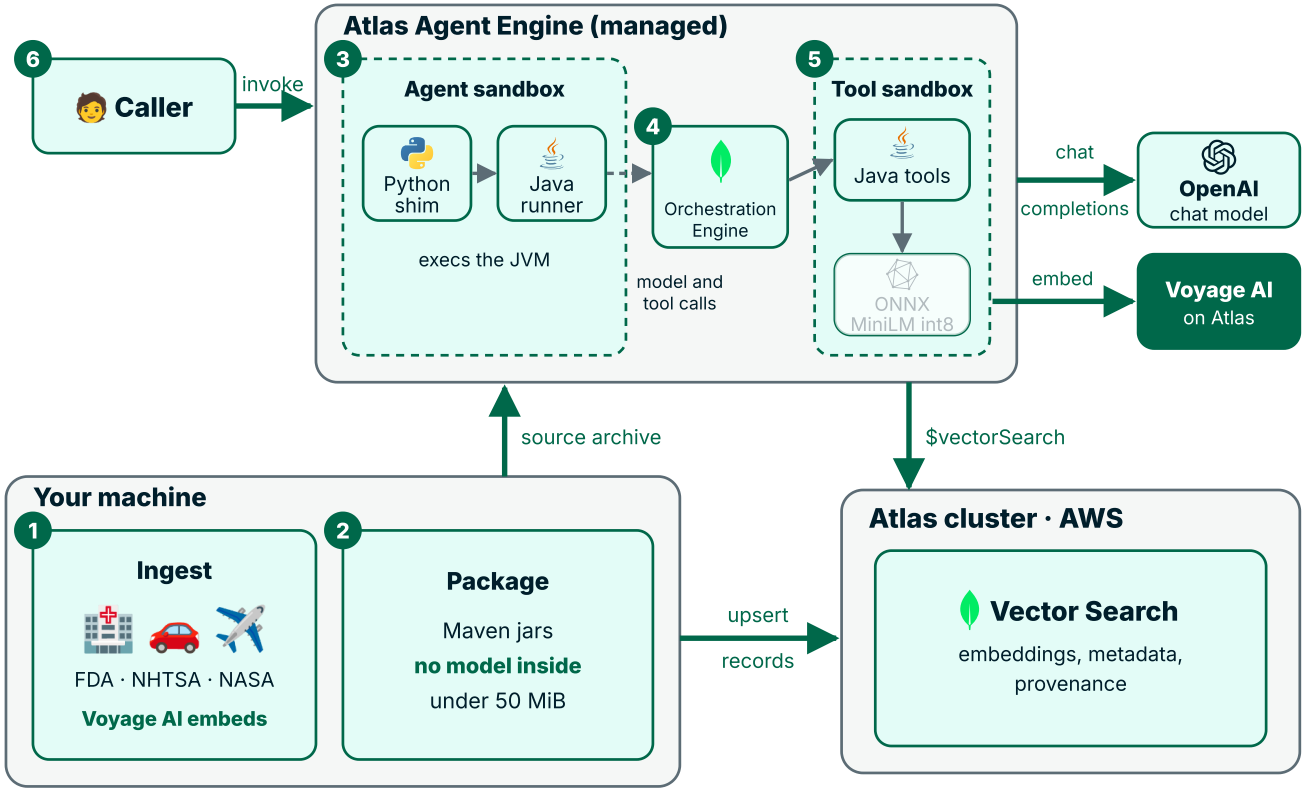
0.479  B737-700, Climb. Air carrier Captain reported an engine
       failure during climb…

0.42   Aeronca Champion, Commuter and GA Icing Incidents…
```

Different words, similar meaning.

Spoiler: the happy ending

"The Road goes ever on and on / Down from the door where it began." — J.R.R. Tolkien, *The Fellowship of the Ring*



1. **Ingest**: embed the public reports and upsert them into Atlas
2. **Package**: bundle the jars and an int8 model under 50 MiB
3. **The stowaway**: start the Java runner from a Python shim
4. **The Orchestration Engine**: approve every model and tool call
5. **Java tools**: embed the query, run the vector search and call the model
6. **The caller**: invoke the agent

✨ **Or flip one switch**: Voyage AI on Atlas does the embedding

CHAPTER 1

Dude, where's my Java SDK?

Introducing the MongoDB Agentic Platform



SHUT UP WITH THE SALES PITCH.

LET ME TRY IT.

⚠️ Atlas Agent Engine is in **public preview**: not yet recommended for production.

Java SDK: 404 ❌

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Sorry, we can't find that page.

The page with the URL "https://www.mongodb.com/docs/agentengine/sdk/java/" does not exist. It might have been moved or deleted.

The MongoDB Assistant suggests the following pages instead:

- Atlas Agent Engine TypeScript SDKs
  https://mongodb.com/docs/agentengine/sdk/javascript
- Table of Contents
  https://mongodb.com/docs/agentengine/sdk/python/packages/agent-engine-sdk-langgraph/docs/api
- @mongodb-js/agent-engine-sdk-langgraph
  https://mongodb.com/docs/agentengine/sdk/javascript/packages/agent-engine-sdk-

worst name ever!

The CLI says no

```
$ agentengine init

Error: unsupported language/framework combination "java-none"

(supported: python-langgraph, python-adk,
            python-openai-agents, typescript-langgraph)
```

This talk is about this error message.

The plan

How hard can it be?

CHAPTER 2

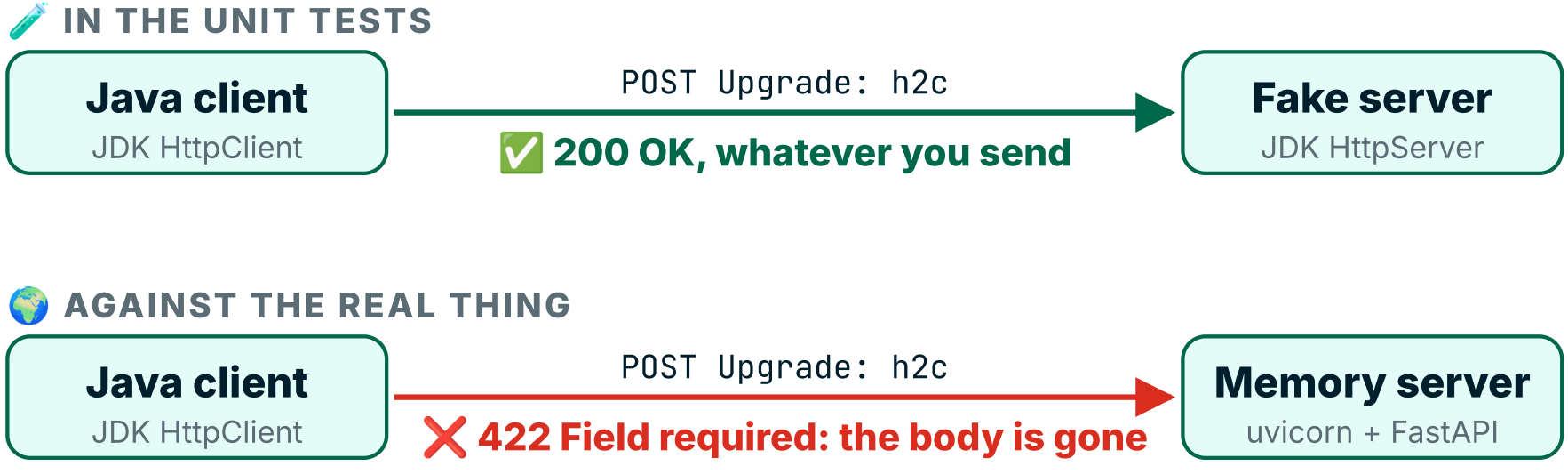
But it works on my machine

core SDK → memory → LangChain4j → runner → LangGraph4j → ☁️

The core SDK

- 📘 The models live in TypeScript (zod schemas)
- ☕ Port them to Java **records**, same wire format
- 🤖 Just **Claude** your way through
- 🟢 19 model tests, green on the first run

Problem #1: Tests pass, writes fail



Why?

☕ **JDK client**: "Can we talk HTTP/2? Also, here's my body."

POST + `Upgrade: h2c` + the memory write

🦄 **uvicorn**: "Upgrade? Body? What body?"

drops the body of the upgraded request

🐍 **FastAPI**: "Field required."

`422` it sees an empty body

🧪 **Fake test server**: "Upgrade? Body? Sure, whatever."

`200` it accepted the upgrade, so every test passed

The fix

Http.java

```
 HttpClient.newBuilder()
+    .version(HttpClient.Version.HTTP_1_1)  // h2c ate the body
     .connectTimeout(connectTimeout)
     .build();
```

MemoryClientTest.java

```
@Test
void plainHttpRequestsDoNotAskForAnH2cUpgrade() {
  client().build().getDistinctDomains(null);
  assertNull(server.only().header("Upgrade"));
}
```

Fake servers only know what you taught them.

Problem #2: Embedding rate limit exceeded

```
searchSemantic

Error: Embedding service rate limit exceeded
```

The fix

Read the log before you theorize.

CHAPTER 3

Tap the Wire

core SDK → memory → **LangChain4j** → runner → LangGraph4j → ☁️

The LangChain4j adapter

- 🔌 Any LangChain4j chat model becomes the platform's LLM
- 🧪 45 offline tests, 6 live against OpenAI
- 🟢 All green. Models: done
- 🏃 Next up: the **runner**

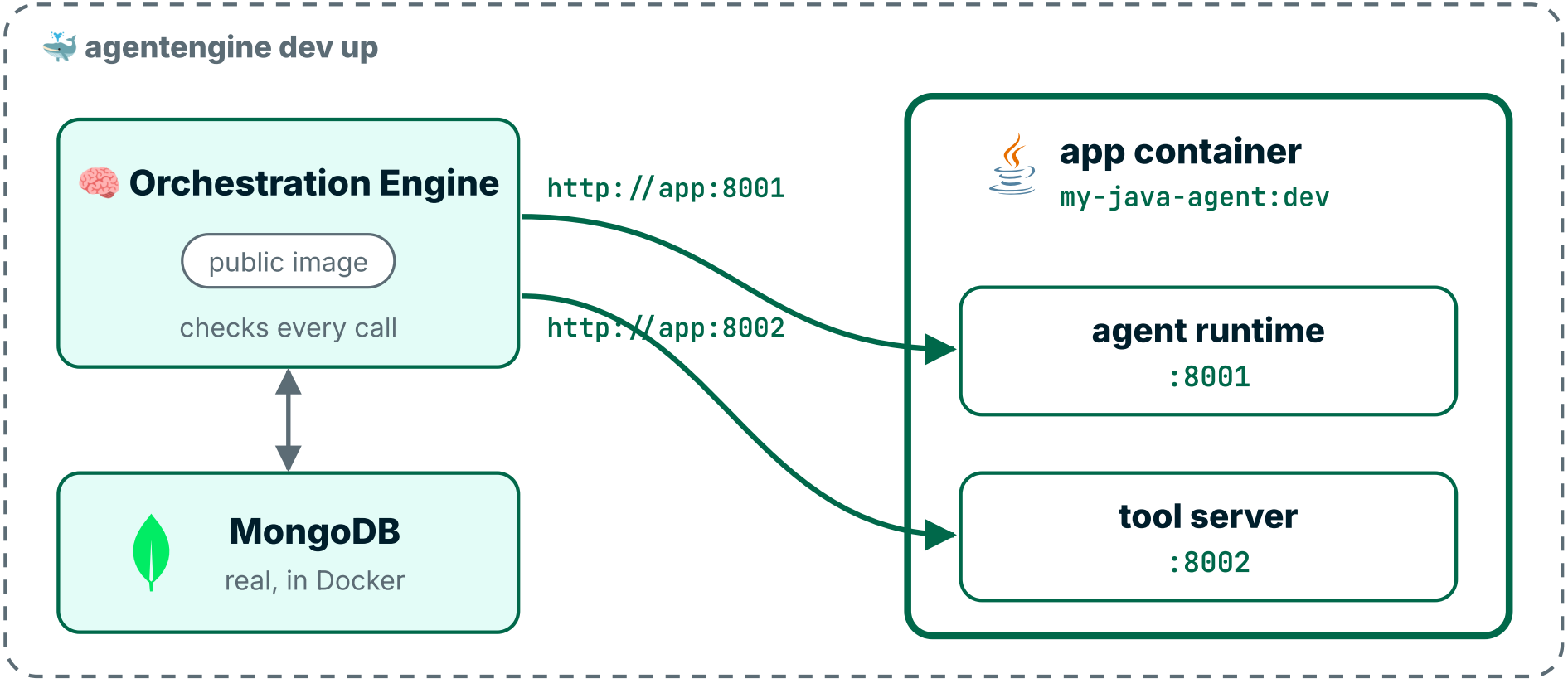
The docs say no Java

core SDK → memory → LangChain4j → **runner** → LangGraph4j → ☁️

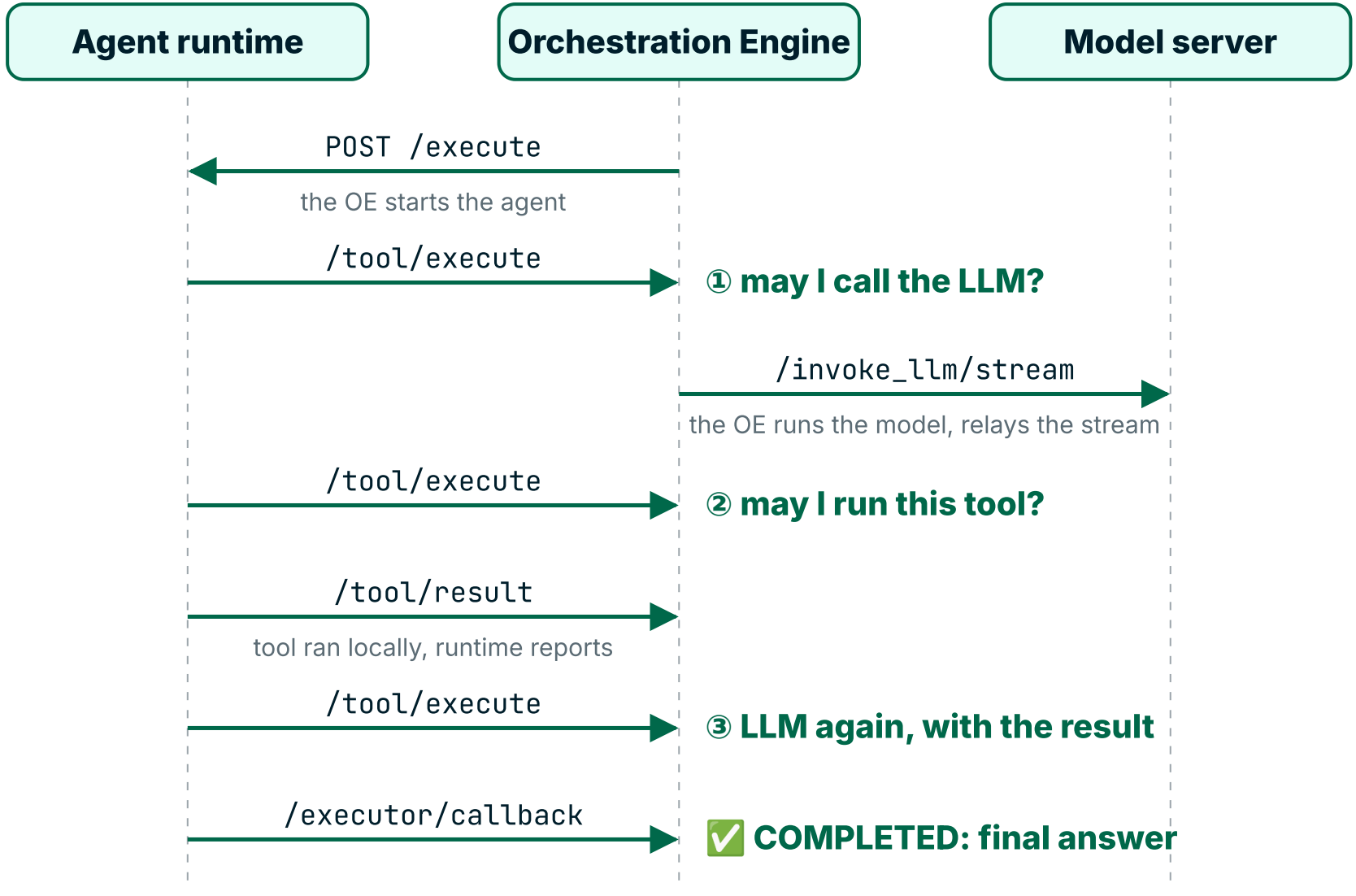
> *"Agents do not implement their own HTTP endpoint."* — *the docs*

So: no Java on the managed platform. 😬

The local stack



The runtime asks first



Will the OE accept a Java container?

☕ **Java runtime**: "Hi. I'm `language=java`."
`POST` /agent/capabilities

🧠 **The OE**: "...Sure. Come in."
`200` OK

```
{
  "result": "Hello ana, from a Java agent!",
  "status": "completed"
}
```

The OE didn't care that I'm Java. 🎉

CHAPTER 4

Graphs, Threads and Amnesia

An agent as a graph

> *"nodes do the work, edges tell what to do next."*
>
> — *LangGraph docs, Graph API › Graphs*

1. 🧵 **Threads**: nodes run on LangGraph4j's own workers
2. 💾 **Checkpoints**: the state saved after every node
3. ⏸️ **Pauses**: a human decides when the graph resumes

Problem #3: No execution on this thread

```
IllegalStateException: No execution is active on this thread
```

The fix

ScopedExecutor.java

```java
// on the thread that has the execution
this.execution = ExecutionScope.current();

// on whichever worker picks the task up
POOL.execute(() ->
    ExecutionScope.call(execution, task));
```

LangGraph4jAgent.java

```
 RunnableConfig.builder().threadId(threadId)
+    .emitterExecutor(new ScopedExecutor())
```

Context tied to a thread stays on that thread.

Problem #4: The agent forgets

🧑 "My name is Ana and I love sailing."

🤖 "Nice to meet you, Ana!"

🧑 "What's today's date? Use your tool."

🤖 "🔧 It's 2026-10-16."

🧑 "What's my name?"

🤷 "I don't know. We haven't met."

The fix

App.java

```
 CompileConfig.builder()
     .checkpointSaver(checkpointer)
+    .releaseThread(false)   // keep it after the run
```

LangGraph4jAgent.java

```
// one session, one thread, every turn
threadId(session, workspaceId)
```

⛵ "Your name is Ana, and you love sailing."

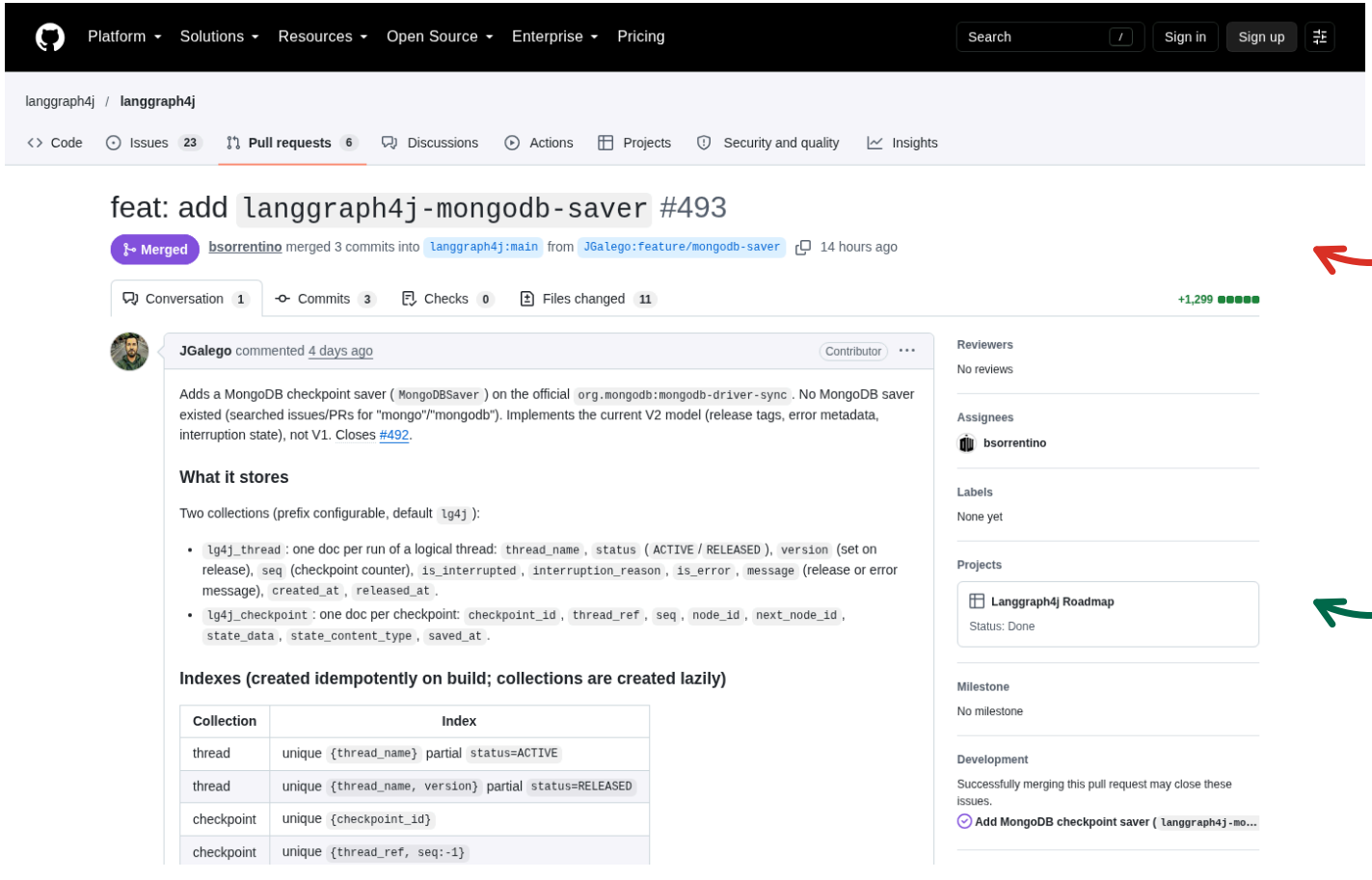
Memory that doesn't last ⚠️

Postgres | Redis | MySQL | Oracle | SQLite

DynamoDB | CockroachDB | Hazelcast | 🍃 MongoDB

Restart the runtime and the agent forgets again.

PR #493: the saver



merged! 🎉

on the roadmap!

Problem #5: The resume decision is nested

```
throw new SuspendException("awaiting_human_review", context);
```

What does the agent receive when it comes back? 🤔

The fix

📄 ApprovalAgent.java

```
-ctx.resumeData().path("decision")
+ctx.resumeData().path("human_review").path("decision")
```

🧑 The reviewer's answer

```
POST /resume/<execution-id>
{"human_review": {"decision": "approve"}}
```

👍 approve → "Approved, going ahead."

👎 reject → "Not approved, nothing was done."

CHAPTER 5

Reports, Not Findings

Problem #6: Every pump query returns boilerplate

```
> insulin pump delivered too much insulin

1. "insufficient information regarding a device problem…"
```

The fix

MaudeIngestor.java

```
// 120 words a chunk, 30 shared, and only the first chunk
TextChunks.split(narrative, 120, 30, 1)
```

Keep the part where someone says what happened.

A report is a claim

🏥 **A report**: "The pump was confirmed defective by medical staff."
what somebody submitted

🤖 **The agent**: "One report states that the pump was defective."
attributed, with the report number

🚫 **Never**: "The pump is defective."
nobody confirmed it

A pacemaker and a Tesla

🧑 "Is there a verified connection between the pacemaker reports and the Tesla Model 3 complaints about sudden loss of power?"

verified_same_* shared report, product code, vehicle, aircraft...

analogous_pattern similar vectors, collected independently

independent similarity below the reporting cutoff

CHAPTER 6

The Stowaway Agent

The wall

```
$ agentengine init

Error: unsupported language/framework combination "java-none"
```

☝️ **Me**: "A hidden option: `AGENTENGINE_IMAGE_SOURCE=custom`. Can I bring my own image?"

🔎 **The template**: "I only mirror the CLI binary and the dev-stack images."
from `agent source template --stdout`

The loophole

🏗️ **The platform**: "I import whatever `agent.yaml` names, check that it's a `BaseApp`, and call `run()`."

```
app = _load_app(config["entrypoint"])   # must be a BaseApp
app.run()
```

Nobody said `run()` has to return.

The stowaway

📄 java_shim/__init__.py

```
class JavaApp(BaseApp):
    def run(self):
        from jdk4py import JAVA           # a JDK... from pip
        os.execv(str(JAVA), [str(JAVA), "-cp", "java/lib/*",
                 "com.mongodb.agentengine.runner.Launcher"])

app = JavaApp("java-agent")
```

1. 🐍 A Python package on paper
2. ☕ A JRE from PyPI inside a wheel
3. 📦 Prebuilt jars in the source tree

Will the build buy it?

```
$ agentengine build logs bld_01M524H4BZG1YJJJQRGJ5NZ6JM

2026-10-16T10:36:50Z Detected agent language: python

2026-10-16T10:37:02Z Using entrypoint: java_shim:app
```

🏗️ **The platform**: "Ah yes, a Python agent." 🐍

Problem #7: The agent never starts

```
$ agentengine invoke "Hello from the managed platform"

    The agent did not become ready during startup.

$ agentengine logs --source agent

    SSLHandshakeException: PKIX path building failed
```

Who's to blame?

🤖 **Claude**: "A private CA. Python trusts it, Java doesn't."
so I wrote a launcher that merged the bundles

🪵 **The log**: "Only the usual public CAs. No private one."
after the redeploy the handshake still failed

🧠 **The OE**: "I want a client certificate."
mutual TLS: the client has to prove who it is, not only the server

🐍 **The Python runner**: "`TLS_CERT_PEM`, `TLS_KEY_PEM`, `TLS_CA_CERT_PEM`. If any of the three is missing I refuse to connect."
the contract was written down only in **`tls_client.py`**

The fix

PlatformTls.java

```
// trust only the platform's CA
trust.setCertificateEntry("ca-" + index++, anchor);

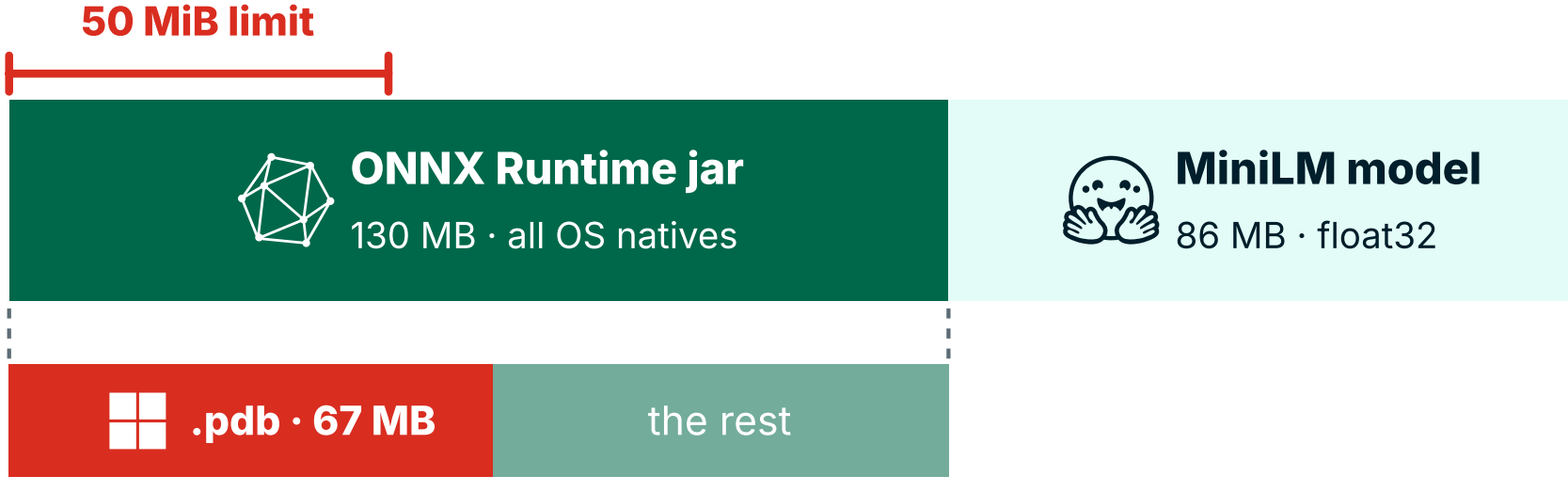
// present the platform's client certificate and key
identity.setKeyEntry("client", privateKey, noPassword, chain);

context.init(keyManagers, trustManagers, null);
```

Built from the three TLS_* variables.
No certificate, no connection.

Problem #8: The package is too big

```
object exceeds 52428800 byte limit
```



The fix

Or ship no model at all

📄 `.env`

```
NEAR_MISS_TEXT_PROVIDER=voyage
VOYAGE_API_KEY=…   # the one memory already uses
```

Problem #8 never happens.

Same key as memory. Nothing new to set up.

Kick the tires

🔑 is the key alive?

```
curl https://eu.ai.mongodb.com/v1/embeddings \
  -H "Authorization: Bearer $VOYAGE_API_KEY" \
  -H "Content-Type: application/json" \
  -d '{"model": "voyage-4", "input_type": "query",
       "input": ["engine lost power during climb"]}'
```

☕ the same demo, embedded in Europe

```
NEAR_MISS_TEXT_PROVIDER=voyage \
VOYAGE_BASE_URL=https://eu.ai.mongodb.com/v1 \
mvn -pl examples/near-miss exec:java -Dexec.args=demo
```

Ask about a car, get a plane

🧑 "car lost power while driving on the highway"

🧮 **MiniLM**, #1: "M-4, Takeoff: engine loss of power inflight" ✈️
a Maule

🚢 **Voyage**, #1: "Honda Accord 2018: blown head gasket" 🚗
three cars

24 questions, no filter. MiniLM crossed domains on 3. Voyage on none.

CHAPTER 7

It's Alive!

And on the real platform?

☕ **Java runtime**: "Hi. I'm `language=java`."
advertised to the managed Orchestration Engine

🧠 **The managed OE**: "...Sure. Come in."
the same answer as on my machine

```
$ agentengine invoke "Hello from the managed platform"

Hello 6ad09241613deba73b039611, from a Java agent!

You said: Hello from the managed platform
```

Only the CLI's language check stood in the way.

Can it find things?

🧑 "Find aviation reports about an engine losing power during climb."

```
1. M-4, Takeoff: engine loss of power inflight    [ACN 2036402](https://

2. B737-700, Climb: engine failure during climb   [ACN 2081933](https://

3. A320, Initial Climb: uncommanded pitch up      [ACN 2032428](https://
```

A Maule loses power at 300 feet



Maule M-4-210C Rocket, Ahunt, public domain

ACN: 2036402 *(12 of 50)*

Component

Aircraft Component : Fuel Booster Pump
Aircraft Reference : X
Problem : Failed

👨‍✈️ **The reporter**: "The fuel pump that was supposed to be removed was in fact still on the plane and the cause of engine loss of power. I have attempted to reach out to seller but am unable to get in touch with him as I believe he is unavailable for next few weeks. Aviation safety at the core is about trust. Trust between pilot, mechanics, crew and all involved. I'm just a pilot, not a mechanic and I expect when a log book is signed airworthy and complied with all ADs and SBs for that to be true."

ASRS report ACN 2036402, narrative

📋 **Primary problem**: aircraft
Cause: according to the reporter

📋 **Contributing factors**: aircraft

Does it make things up?

🧑 "Is there a verified connection between the pacemaker reports and the Tesla Model 3 complaints about sudden loss of power?"

```
There is no verified connection. The FDA MAUDE and NHTSA
records do not share provenance, so they are not linked by
the same report or product.
```

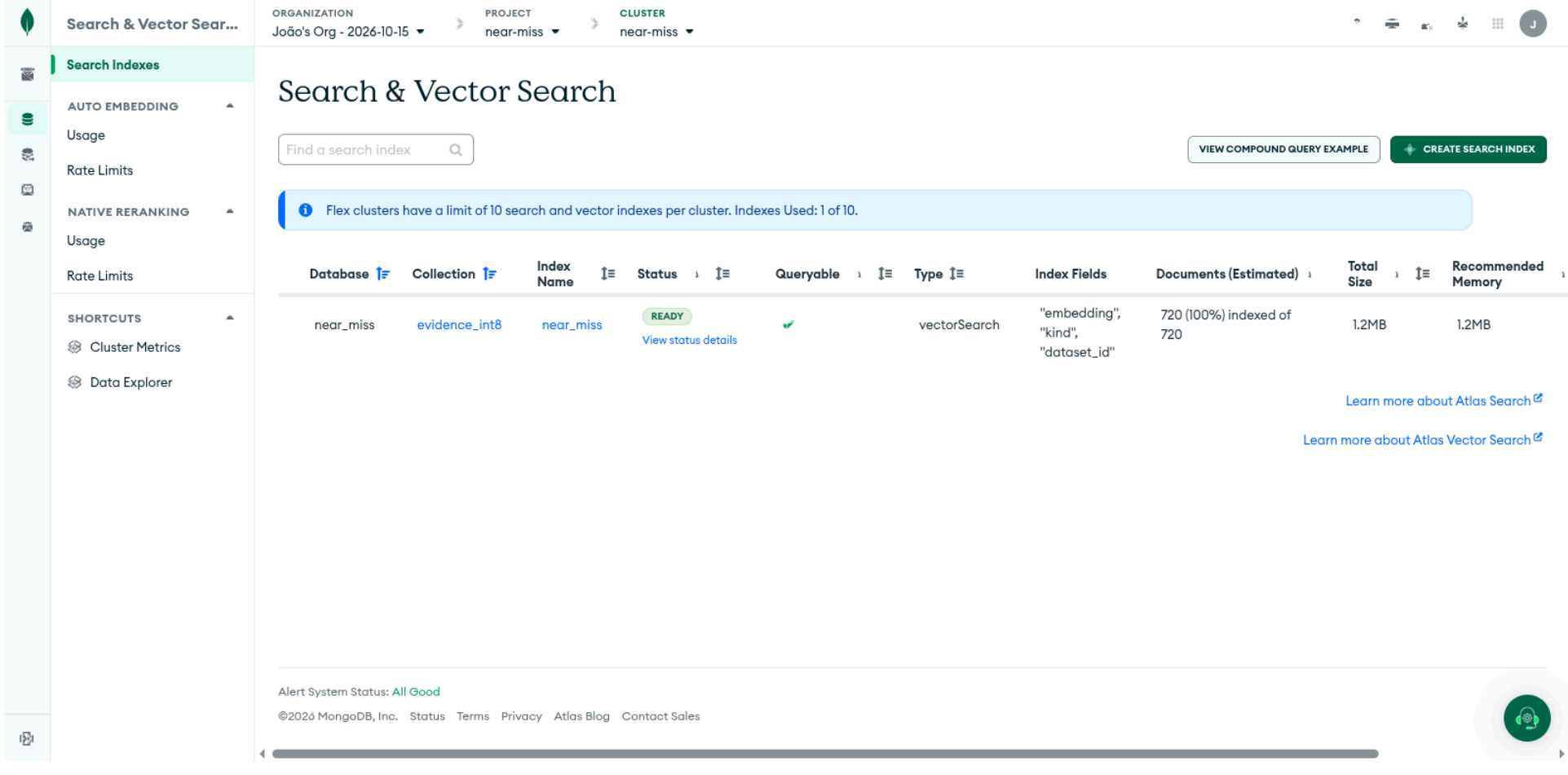
Does it give advice?

🧑 "Is the Tandem Mobi insulin pump defective? Should I stop using mine?"

```
One report (MW5176973) states that the original pump was confirmed
defective by medical staff […]

For concerns about whether to continue using your pump, consult
the manufacturer, a clinician, or the agency that took the report.
```

The vector index is READY



What we learned along the way

🪵 **Read the log before you theorize** · Problem #2

🧪 **Your fake server may be lying to you** · Problem #1

🐍 **Read the source when the docs are silent** · Problem #7

🔬 **Probe, don't guess** · Problem #9

📝 **Every workaround is a feature request** · the next slide



16 October 2026

Dear MongoDB Atlas Agent Engine Team,

I'm writing to you from the other side of a 404, and I'm happy to report that a Java agent now runs on Atlas Agent Engine. Your Orchestration Engine took one look at it and said "come in".

I should admit how I got there. I wrote a small Python shim that boots a JVM inside your runner, and when the docs went quiet I read your code. Sorry about that. It works, but one platform update could break the whole thing.

So I have a small request: could `agentengine init` learn to say `java`? A Java runner image, a documented runner contract or simply bring-your-own-image would each do the job, and any of them would turn my stowaway into a paying passenger.

Thank you for building something worth sneaking into.

All the best,

João Galego

P.S. No hard feelings about `java-none`.

```
@@ -1 +1 @@
- Java vs. MongoDB
+ Java + MongoDB
```

Thanks!



JGalego/agent-engine-client-libraries

`feat/java-support`

Scan it, deploy it, tell me what breaks.